MCA101 – Probability, Statistics & Computational Mathematics

Written Assignment

(Answer all the questions in your notebook and upload the same in Moodle in Pdf format before the deadline.)

1. Let p stand for the proposition “I bought a lottery ticket” and q for “I won the jackpot”. Express the following as natural English sentences:

a) $p \wedge q$
b) $p \Rightarrow q$
c) $\neg p \Rightarrow \neg q$

2. Compute the truth table of $(F \vee G) \wedge \neg(F \wedge G)$.

3. Use the truth tables method to determine whether $p \rightarrow (q \wedge \neg q)$ and $\neg p$ are logically equivalent.

4. In a group of 200 students, 110 study AI, 90 study Data Analytics, and 60 study Cybersecurity. The pairwise intersections are 40, 35, and 30, respectively, and 20 study all three. Find the number studying at least one specialization.

5. In a class of 120 students, 65 study Python, 55 study Java, and 40 study C++. Also, 25 study Python and Java, 20 study Python and C++, 15 study Java and C++, and 10 study all three.

a) How many study at least one language?
b) How many students study none of the three languages?
c) How many students study exactly two languages?